FORM 2

THE PATENTS ACT, 1970
(39 of 1970)
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COMPLETE SPECIFICATION

(See section 10; rule 13)

1. TITLE OF THE INVENTION

"HYBRID DEEP LEARNING BASED METHOD AND SYSTEM FOR AQUACULTURE WATER QUALITY PREDICTION"

2. APPLICANT (S)

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3. PREAMBLE TO THE DESCRIPTION

The following specification particularly describes the invention and the manner in which it is to be performed.

HYBRID DEEP LEARNING BASED METHOD AND SYSTEM FOR AQUACULTURE WATER QUALITY PREDICTION

FIELD OF THE INVENTION

The present invention relates to the field of water quality management in aquaculture. Particularly, the present invention relates to a method and system for predicting water quality in aquaculture using deep learning techniques, specifically a hybrid model combining convolutional neural networks (CNNs) with long short-term memory (LSTM) or gated recurrent unit (GRU) networks. The invention is intended to improve the efficiency and effectiveness of aquaculture operations by providing early warning of potential water quality issues, enabling timely corrective action to be taken.

BACKGROUND OF THE INVENTION

Aquaculture is vital for global food security as the world population is projected to reach 9.8 billion by 2050. Natural resources are at their limit and the ocean alone may not suffice to meet the demand for food. Aquaculture is an efficient solution for food production, yet its environmental impact needs to be assessed. It is estimated that by 2050, aquaculture will produce one-third of all seafood. Sustainable aquaculture practices are being adopted to ensure the production of healthy and safe food while preserving the environment. Aquaculture, the farming of aquatic organisms such as fish, shellfish, and seaweed, is a growing industry that plays a vital role in meeting the increasing global demand for seafood. However, the success of aquaculture operations depends on many factors, including water quality. Poor water quality can lead to reduced growth and survival rates of the cultured species, as well as increased susceptibility to diseases. Aquaculture is considered as an integral component for global food security as the world population is expected to reach 9.8 billion by 2050. The harvest from natural resources is mostly beyond their biological limits, and the ocean alone may not meet the demand for food required for this growing population.

Aquaculture is one of the efficient solutions for food production but not without impacting the environment. Aquaculture involves inefficient energy use, large carbon footprint and waste of water resources. To minimize the impact on the environment and optimize the use of resources, Water quality prediction (WQP) plays an important role.

Usually, the farmer assesses the farm by continuous manual monitoring, which will not provide all the required information. This traditional method is tedious, time-consuming, and incapable of meeting the requirements. They can take action only when the farmer has discovered a visible change in the condition of the fish or environment. By this time, environmental factors have already changed dramatically. The procedure to bring the farming environment back to normal is expensive once the damage is done. There are also other traditional methods of water quality monitoring where the water samples are collected and sent to a lab. This method consumes time, and they delay the decision-making process affecting productivity. But the same is done by using sensors that monitor the physical environment continuously and provide us with extensive data which may include all the factors which farmer cannot assess manually. This may help us to control the system and reduce the impact on the environment. WQM may also help us with enhancing the quality of the environment, improving product quality, and increasing production. Thus, monitoring of the fish farm optimizes the use of resources improving sustainability and may help the farmer make a better profit.

The vital water quality parameters in aquaculture are temperature, dissolved oxygen (DO), nitrate, ammonia, turbidity, salinity, pH level and alkalinity. The requirement varies for different species, and farmers need to be aware of them. The fluctuation in these parameters may result in massive production loss. Several reasons are there for contamination of water. Even seawater may be polluted due to urbanization and industrialization. Dissolved Oxygen (DO) is one of the critical water quality

parameters, which reflects changes in water quality in aquaculture. The water quality of an aquaculture pond has a direct impact on the growth of aquatic animals and product quality. These parameters directly affect fish health and feed utilization affecting the growth rate. Fish experience stress and diseases breakout when the temperature is chronically near their maximum tolerance or fluctuates suddenly.

The water quality management of farmers is based on assumption and is done periodically without any assessment. In some cases, they will replace the water once every three days without considering the requirements and the source may be even more polluted. Some others may use different chemicals periodically. All these practices without proper water quality monitoring will lead to mismanagement and pollution of natural resources. The farmers are never using any monitoring techniques for water quality. A few of them are periodically using different manual testing kits to check water parameters. Others are even evaluating fish behaviour, which creates stress and affects growth. One segment of traditional farmers is unaware of technology and not affordable for many who are aware. The majority production in aquaculture comes from small farms and helping them to be sustainable and efficient farmers will have a considerable impact on the overall production output and reduce the impact of aquaculture on the environment.

To maintain and manage aquaculture, water quality monitoring and water quality prediction (WQP) are essential. In aquaculture, water quality is the most influential and critical factor that affects production as well as product quality. The vital water quality parameters in aquaculture are salinity, pH, dissolved oxygen (DO), and temperature. Water quality is affected by many factors, such as fish density, quality of the feed, feeding interval, climate, and more. The change in water quality will upset the balance of the system with algae bloom, bacterial growth, etc. This can lead to severe problems, such as trigger stress, lack of food intake, vulnerability to diseases, and increased mortality rate of fish. Hence, if the trend in water quality

changes is forecasted, safeguards can be employed in advance to avoid imbalances in the ecosystem and ensure suitable conditions for optimum growth of the fish. Therefore, accurate WQP can drastically improve productivity to make aquaculture more profitable and sustainable.

Aquaculture water quality is affected by meteorological conditions and complex interdependency relations between different water quality parameters. Hence the change of water quality parameters exhibits non-linear characteristics, which results in low prediction accuracy. The time-series data shows periodic variations depending on the seasons and climatic conditions. Time series data is used for analysis and prediction in various fields such as the stock market, medical field, energy consumption, weather forecasting, solar radiation etc. The major approaches for WQP include classical prediction mechanisms such as the autoregressive moving average model (ARMA), autoregressive integrated moving average (ARIMA), seasonal autoregressive integrated moving average model (SARIMA), seasonal autoregressive integrated moving average with exogenous factors (SARIMAX), Holt-Winters exponential smoothing (HWES), Markov model, Grey model, and support vector regression (SVR) etc. However, these models are least suitable for aquaculture WQP because they assume a linear relationship with water quality parameter data. These models are less accurate and take a long prediction time, making them unsuitable for predicting the non-linear aquaculture water quality parameters.

Deep learning (DL) models like long short-term memory (LSTM), gated recurrent unit (GRU), and convolutional neural network (CNN) have the flexibility in capturing the non-linear nature of the aquaculture WQP. These DL models have overcome the limitations of Recurrent neural networks (RNNs) and achieved great success in various applications. Recent advancements in artificial intelligence have made DL one of the most proficient methods, and DL is popularly applied in different fields such as image processing, speech recognition, text prediction, etc. DL makes time

series prediction more precise and efficient in terms of training time and required processing power. In addition, DL models consider the non-linear characteristics of water quality changes. LSTM is the widely used DL method for WQP due to its remarkable performance in time series prediction.

LSTM and GRU models are not very capable of maintaining long-term memory for time series prediction, especially for long sequences. In time series forecasting, the prediction value at the current time step is impacted mainly by past observations. In some cases, the ones with a significant influence might have appeared at the time step long back from the current one. Hence neural networks must have the prolonged memory holding capacity to hold memory for the learned long-term dependencies. The ability of LSTM models to capture long-term dependencies information from historical observations is still considered a critical performance bottleneck, as indicated in recent studies. The work has theoretically proven that standard LSTM does not have long memory from a statistical perspective. Hence, building a time-series prediction model capable of capturing and remembering complex dependencies is a crucial problem that needs to be addressed.

The hybrid CNN-LSTM models also work similarly, helping the LSTM model capture the relevant data. In hybrid models, CNN with maxpooling is used to reduce the length of the input sequence, which is fed into the LSTM and GRU models. The hybrid models are not affected by exponential decay, as the CNN with maxpooling will input the key features and helps LSTM in forecasting with better accuracy. Thus, the hybrid models can achieve higher accuracy in prediction. Hybrid models aim to build neither overfit nor underfit models with strong generalization ability and require less training time for fitting the models. The convolutional layer with maxpooling enhances the capacity of the LSTM and GRU models to learn and store the critical features in the non-linear water quality parameters. Thus, convolution layer with

maxpooling reduces LSTM and GRU's exponential decay, bypassing selective information to the LSTM/GRU layer using convolution and maxpooling.

CNN and its hybrid variations are used in many research work for time series prediction. DL models have been extensively used in many fields of time series prediction, including finance (stock price, cryptocurrency, and precious metal prediction). The authors have used the CNN-LSTM model for gold price prediction. They have proposed two simple hybrid models and compared the results with other DL models. The authors also compared LSTM models with the hybrid models and observed that hybrid models perform better. In to predict blood glucose (BG) levels, authors have used multiple layers of CNN along with LSTM. This hybrid model has shown superior performance in BG prediction. In another work, the authors have used Bi-LSTM with CNN for air quality prediction. They have done an extensive study with two datasets and shown results proving the model's capability for accurate air quality prediction. There have been multiple works based on hybrid DL models for the prediction of air quality (PM2.5) based on CNN and LSTM models.

Water quality prediction (WQP) and Water quality prediction (WQP) are important aspects of aquaculture, as it ensures the health and safety of aquatic organisms. It is essential to track water properties such as temperature, pH, dissolved oxygen (DO), salinity and turbidity to detect changes in water quality. Therefore, it is important to monitor and manage water quality in aquaculture systems to ensure optimal conditions for the cultured organisms.

Traditionally, water quality in aquaculture systems has been monitored using various sensors and instruments, such as pH meters, dissolved oxygen meters, and temperature sensors. These instruments can provide real-time data on various water quality parameters, such as pH, dissolved oxygen, temperature, salinity, and nutrients. However, these measurements are often taken at fixed intervals and do not provide a

comprehensive picture of water quality over time. Furthermore, interpreting and analyzing the data collected from these sensors can be challenging, as it requires expertise in both aquaculture and water quality management.

In recent years, there has been increasing interest in using deep learning techniques to improve the monitoring and management of water quality in aquaculture systems. Deep learning is a type of machine learning that involves training artificial neural networks on large datasets to learn patterns and make predictions. Convolutional neural networks (CNNs) are a type of deep learning model that are particularly well-suited for image analysis tasks, such as classifying and detecting objects in images. Long short-term memory (LSTM) and gated recurrent unit (GRU) networks are types of deep learning models that are specifically designed for processing sequential data, such as time series data.

The combination of CNNs and LSTM/GRU networks, referred to as a hybrid model, has been shown to be effective for various prediction tasks, including water quality prediction. In a hybrid model, the CNN is used to extract features from the data, while the LSTM/GRU network is used to analyze the extracted features over time to make predictions.

CN110334845A discusses cultivation of steamed crab and GRU based dissolved oxygen long-time prediction method. Here GRU is used for predicting dissolved oxygen. The accuracy and computational efficiency are less, as this work uses only GRU and it's difficult for GRU to fetch the nonlinear characteristics of water quality.

CN114881303A discusses water quality long-term prediction method based on double attention mechanism which helps in improving the accuracy of the water quality prediction. In this work the important features are extracted using the attention mechanism helps in improving the accuracy of water quality prediction. But the

attention based models require extra computation power when compared with the CNN based hybrid deep learning models. This eliminates the scope of implementing this model in any edge device and the water quality prediction system either requires power hungry workstations or must depend on the cloud which attracts subscription charges making the system expensive in terms of energy and cost.

NPL titled "Prediction of Water Level and Water Quality Using a CNN-LSTM Combined Deep Learning Approach" discusses a CNN model used for predicting the water level and LSTM model is used for predicting different water quality parameters. Two different models are used for predicting the water level and the water quality, which is totally different from using hybrid models combining their performance for water quality prediction. The CNN model is used to predict the water level, and the non-linear water quality parameter is predicted using LSTM model. But this doesn't help to reduce computational efficiency or improve the accuracy of water quality prediction. In this work water quality including total nitrogen, total phosphorous, and total organic carbon along with water level is predicted.

NPL titled "A Long-Term Water Quality Prediction Method Based on the Temporal Convolutional Network in Smart Mariculture" discusses the water quality prediction method based on TCN (temporal convolutional network), and predicts dissolved oxygen, water temperature, and pH. Long-term prediction method based on TCN has higher accuracy and less time complexity, compared with RNN (recurrent neural network), SRU (simple recurrent unit), BI-SRU (bi-directional simple recurrent unit) and LSTM (long short-term memory). These two models are not compared directly, but both are compared to basic models and have superior performance compared to that. The TCN prediction network extract time series features and in-depth data features by introducing dilated causal convolution and has a good effect of long-term prediction.

NPL titled “Short-term water quality variable prediction using a hybrid CNN–LSTM deep learning model” discusses the development of two standalone deep learning models, long short-term memory (LSTM) and convolutional neural network (CNN), and a hybrid CNN–LSTM model to predict two water quality variables (dissolved oxygen (DO) and chlorophyll-a (Chl-a)) in the Small Prespa Lake in Greece. In this work they only have developed hybrid CNN-LSTM model and are compared with two traditional machine learning models, support-vector regression (SVR) and decision tree (DT). This hybrid CNN-LSTM model have similar and computationally efficient, but the CNN-GRU model is further efficient when compared to the CNN-LSTM model.

The difference between the two models is the application of GRU in the second layer, in this work we only use the LSTM as second layer, whereas in the proposed model both LSTM and GRU are used, and it helps in improving its performance. However, the success of marine aquaculture operations depends on many factors, including water quality.

The prior arts discussed herein don’t have hybrid CNN-LSTM/GRU model rather using the normal deep learning models. Hence, there is a need to overcome the above-mentioned problem and there is a need for a system to monitor and manage water quality in marine aquaculture systems, providing early warning of potential water quality issues and taking timely corrective action.

OBJECTS OF THE INVENTION

The primary object of the present invention is to provide a method and system for predicting water quality in aquaculture systems that is based on deep learning techniques, specifically a hybrid model combining convolutional neural networks (CNNs) with long short-term memory (LSTM) or gated recurrent unit (GRU) networks.

Another object of the present invention is to provide a method and system for predicting water quality in aquaculture systems that is able to analyze data from various sensors and instruments in real-time and uses this data to make predictions about the water quality in the aquaculture system.

Yet another object of the present invention is to provide a method and system for predicting water quality in aquaculture systems that is user-friendly and can be easily integrated into existing aquaculture management systems.

Yet another object of the present invention is to provide a method and system for predicting water quality in aquaculture systems that is able to provide early warning of potential water quality issues, enabling timely corrective action to be taken.

Yet another object of the present invention is to improve the efficiency and effectiveness of aquaculture operations by providing a method and system for predicting water quality in aquaculture systems.

Yet another object of the present invention is to reduce the risk of water quality issues in aquaculture systems.

SUMMARY OF THE INVENTION

One or more of the problems of the conventional prior art are overcome by various embodiments of the present invention.

A system for predicting the water quality in aquaculture systems using a hybrid model that combines convolutional neural networks (CNNs) with long short-term memory (LSTM) or gated recurrent unit (GRU) networks. The system is designed to analyze data from various sensors and instruments in real-time and uses this data to make predictions about the water quality in the aquaculture system.

The goal of the system is to improve the efficiency and effectiveness of aquaculture operations by providing early warning of potential water quality issues, enabling timely corrective action to be taken. The system is intended to be user-friendly and can be easily integrated into existing aquaculture management systems. The system can be used in various types of aquaculture systems, including but not limited to marine, freshwater, and brackish water systems.

It is primary aspect of the present invention to provide a water quality prediction (WQP) system using deep learning techniques, comprising:

one or more nodal units comprises of two or more sensor nodes S, and a central node C; and

a water medium,

wherein the central node C is connected to N number of sensor nodes S, each sensor node S comprises of a transceiver, a microcontroller unit and an antenna with two or more sensors, the central node C comprises of a transceiver, a microcontroller unit, a display unit, a Global System for Mobile Communications (GSM) modem, an user device, a server and an antenna,

wherein the sensor node S is deployed in the water medium to acquire water quality parameters, and the estimation of water quality parameter values and environmental parameter values is calculated and the sensed signal is converted to respective water quality parameters, the microcontroller of the central node C is provided for controlling and receiving data from the sensor node S, and the value of water quality parameter acquired electrical signals by the physical sensors, and wherein the microcontrollers of the nodal units at the sensor node S is embedded with instructions.

It is another aspect of the present invention, wherein the sensors acquire environmental parameters such as temperature, pH, salinity, dissolved oxygen and the like.

It is another aspect of the present invention, wherein the water quality parameters comprise of dissolved oxygen, salinity and the like.

It is another aspect of the present invention, wherein the microcontroller unit of the nodal units comprises of a memory module and an input/output module.

It is another aspect of the present invention, wherein the memory module is configured to store the water quality parameters and the input/output module is connected to the sensors configured to receive the sensing signals.

It is another aspect of the present invention, wherein the water quality parameters are sent to the central node C from the sensor node S using the transceiver and the antenna, received at the central node C and stored in a memory and the central node C communicates the information and uploads the water quality parameters to the server.

It is another aspect of the present invention, wherein the server is a cloud server.

It is another aspect of the present invention, wherein the microcontroller is configured to use the GSM modem to send water quality parameters information to the user device by the cloud server if online and by the GSM modem if offline.

It is another aspect of the present invention, wherein the user device comprises of one or more laptops, mobile, smart devices or any other device.

It is another aspect of the present invention to provide a method of working of the water quality prediction (WQP), comprising:

initialization of the sensors for water quality parameters in the water medium and output the sensed signal;

reading the sensors and storing the values in the memory by the microcontroller;

processing the raw data from each sensor and converts to the water quality parameters values in the microcontroller;

estimation of water quality parameters;

sending data to the central node C;

waiting for predefined time;

receiving data from the sensor node S;

storing the data in the memory;

analyzation of the data;

checking whether the parameters are in normal range;

wherein if the parameters are in normal range, the information is communicated to the user and the data is displayed to the user device by the display unit,

wherein if the parameters are in abnormal range, the information is communicated to the user by an alarm unit,

waiting for predefined time;

wherein the operations is to determine to alarm the user or pass information in a defined time interval, and

displaying the water quality parameters values to the user device by the display unit,

wherein the process is repeated continuously passing the information to the end-user and uploading the data to the cloud server.

BRIEF DESCRIPTION OF THE DRAWINGS

The manner in which the features, advantages and objects of the invention, as well as others which will become apparent, may be understood in more detail, more particular description of the invention briefly summarized above may be had by reference to the embodiment thereof which is illustrated in the appended drawings, which form a part of this specification. It is to be noted, however, that the drawings illustrate only a preferred embodiment of the invention and is therefore not to be considered limiting of the invention's scope as it may admit to other equally effective embodiments.

Fig. 1 illustrates a block diagram of the system for water quality prediction (WQP) with central node and 'N' number of sensor nodes according to one embodiment of the present invention.

Fig. 2 illustrates the block diagram of the sensor node of water quality prediction (WQP) according to another embodiment of the present invention.

Fig. 3 illustrates the block diagram of the central node of the system for water quality prediction (WQP) according to another embodiment of the present invention.

Fig. 4 illustrates the flow chart including the method of sensor initialization, data collection, estimation of parameters from physical sensors and communicates the same with the Node C in accordance with an embodiment of the present invention.

Fig. 5 illustrates the flow chart of method of receiving the sensed data at Node C, storing data, uploading data to cloud, analyzing data, and prediction of future values using deep learning hybrid CNN-LSTM/GRU model, in accordance with an embodiment of the present invention.

NAME OF THE APPLICANT: NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY

DESCRIPTION FOR DRAWINGS WITH REFERENCE NUMERALS:

1. Water quality prediction system (100)
2. Water medium (110)
3. Central node C (120)
4. Sensor nodes S (130a…….130n)
5. Microcontroller unit (210, 310)
6. Transceiver (220, 320)
7. Antenna (230, 330)
8. Sensors (240a, 240b, 240c, 240d)
9. Display unit (340)
10. Global System for Mobile Communications (GSM) modem (350)
11. User device (360)
12. Server (370)

DETAILED DESCRIPTION OF THE INVENTION

It is to be understood that the present disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. This particular form is only an exemplary embodiment and is not intended to be taken restrictively to imply any limitation on the scope of the present invention. The present disclosure is capable of other embodiments and of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting.

The present invention relates to the field of water quality management in aquaculture. Particularly, the present invention relates to a method and system for predicting water quality in aquaculture using deep learning techniques, specifically a hybrid model combining convolutional neural networks (CNNs) with long short-term memory (LSTM) or gated recurrent unit (GRU) networks. The invention is intended to improve the efficiency and effectiveness of aquaculture operations by providing early warning of potential water quality issues, enabling timely corrective action to be taken. This technology can be used in various types of aquaculture systems, including but not limited to marine, freshwater, and brackish water systems.

The present invention relates to a method and system for predicting water quality in aquaculture using a hybrid deep learning approach. Aquaculture, also known as fish farming, is the cultivation of aquatic animals and plants in controlled environments, such as ponds, tanks, and raceways. Water quality is a critical factor in the success of aquaculture operations, as it affects the growth and health of the cultured organisms. Poor water quality can lead to diseases and mortality, which can result in significant economic losses for aquaculture farmers.

The proposed method involves the use of a hybrid deep learning model, which combines a convolutional neural network (CNN) with a long short-term memory (LSTM) or gated recurrent unit (GRU) network. CNN is a type of neural network that is particularly well-suited for processing data with a grid-like structure, such as images. The LSTM and GRU are types of recurrent neural networks (RNNs) that are designed to process sequential data, such as time series.

To build the hybrid deep learning model, data from various sensors installed in the aquaculture system is collected and processed. The data includes historical water quality data and corresponding environmental parameters, such as temperature, pH, salinity and dissolved oxygen. This data is used to train and optimize the hybrid deep learning model.

Once trained, the hybrid deep learning model can be used to predict water quality indicators, such as temperature, pH, salinity, and dissolved oxygen. These predictions are made using data from the sensors installed in the aquaculture system and can be used to make timely interventions and decision making in the management of the system.

The proposed method and system demonstrate improved accuracy and robustness in water quality prediction compared to traditional machine learning approaches. This is because the hybrid deep learning model can effectively analyze and extract relevant features from the data, allowing for more accurate predictions. Additionally, the hybrid approach allows for the integration of both spatial and temporal information, which is crucial for water quality prediction in aquaculture systems.

The microcontroller converts these sensed electrical signals to corresponding water quality parameters. The memory module stores a plurality of sensor values to the respective water quality parameters. After estimating all the parameters, the data is

sent to the central node by the microcontroller. The central node collects this data stores them in the database, pass this information to the user in defined intervals. The central node also determines whether the parameters are in abnormal ranges and issues warning signal according to the determined result.

The water quality prediction system for aquaculture consists of Node C for controlling and receiving the data from many Node S, which have the sensors for sensing water quality parameters. All the water quality data is stored in cloud and using CNN-LSTM/GRU deep learning model future values of water quality is predicted using cloud/edge computing in the Node C. The water quality prediction system includes a set of sensors and a microcontroller system. The sensors are used to sense the water quality parameters and output a set of sensing signals that correspond to the water quality parameters.

The microcontroller system includes a memory module, and an input/output module. The memory module is used to store the water quality parameters. The input/output module connected to the sensors is used to receive the sensing signals. The microcontroller converts all sensed signal to respective water quality parameters. The water quality parameters are sent to Node C from Node S using the transceiver and antenna, received at Node C and stored in its memory. Further data analysis and water quality prediction (WQP) is done at Node C. Node C communicates the information and uploads the water quality parameters to the cloud.

Fig. 1 illustrates a water quality prediction (WQP) system (100) with Node C (120) and ‘N’ number of Node S. The Node S (130) may be deployed around the Node C (120) and may choose any mode of communication with the Node C. Multiple Node S (130) is deployed in the waterbody (110) to acquire water quality parameters from different areas of the water body. The aquaculture water quality prediction system (100) consists of a Node C (120) as center point and a set of Node S(130a-n) for

sensing water quality parameters and reporting back to Node C. WQP is implemented in the cloud and accessed via Node C, where DL is used to predict water quality using the data monitored using the WQP system (100).

The system (100) for WQP, comprises one or more nodal units comprises of two or more sensor nodes S (130), and a central node C (120), and a water medium (110). The central node C (120) is connected to N number of sensor nodes S (130). Each sensor node S (130a.....130n) comprises of a transceiver (220), a microcontroller unit (210) and an antenna (230) with two or more sensors (240a, 240b, 240c, 240d).

The central node C (120) comprises of a transceiver (320), a microcontroller unit (310), a display unit (340), a Global System for Mobile Communications (GSM) modem (350), a user device (360), a server (370) and an antenna (330). The sensor node S (130) is deployed in the water medium (110) to acquire water quality parameters (240a, 240b, 240c, 240d), and the estimation of water quality parameter values (240d, 240e) and environmental parameter values is calculated, and the sensed signal is converted to respective water quality parameters (240d, 240e) by the microcontroller (210) of the sensor node S (130).

In an embodiment, the microcontroller (310) of the central node C (120) is provided for controlling and receiving data from the sensor node S (130), and the value of water quality parameter values is sensed from acquired electrical signals by the physical sensors (240a, 240b, 240c, 240d). The microcontrollers (210, 310) of the nodal units (130, 120) at the sensor node S is embedded with instructions. The sensors acquire environmental parameters such as temperature (240a), pH (240b), salinity (240c), dissolved oxygen (240d) and the like.

In an embodiment, the microcontroller unit (210, 310) of the nodal units (130, 120) comprises of a memory module and an input/output module. The memory module is

configured to store the water quality parameters (240a, 240b, 240c, 240d), the input/output module is connected to the sensors (240a, 240b, 240c, 240d) configured to receive the sensing signals, using physical sensors. The water quality parameters (240a, 240b, 240c, 240d) are sent to the central node C (120) from the sensor node S (130) using the transceiver (220, 320) and the antenna (230, 330), received at the central node C (120) and stored in a memory and the central node C (120) communicates the information and uploads the water quality parameters (240a, 240b, 240c, 240d) to the server (310). The server (310) is a cloud server.

The microcontroller is configured to use the GSM modem (350) to send water quality parameters (240a, 240b, 240c, 240d) information to the user device (360) by the cloud server (310) if online and by the GSM modem (350) if offline. The user device (360) comprises of one or more laptops, mobile, smart devices, or any other device.

Fig. 2 illustrates various parts of Node S (200) that are used in the WQP system (100). The Node S (200) includes a microcontroller (210), a transceiver (220), an antenna (230) with multiple sensors (240a-240d). Sensors (240a-240d) acquire environmental parameters such as temperature sensor (240a), pH sensor (240b) salinity sensor(240c) and dissolved oxygen sensor(240d). The input/output module is connected to sensors (240a-240d) and is configured to receive a plurality of sensing signals that correspond to the water quality parameters.

The microcontroller (210) processes the signal received from sensors (240a-240d) through the input/output module and converts available raw signals to water quality parameters values. All the calculations on raw sensor data are done in microcontroller (210) and the processed data is passed to Node C (300) as water quality parameter.

Fig. 3 illustrates various components which may be utilized in Node C (300) that may be employed within the water quality prediction system (100). The Node C (300)

includes a microcontroller (310), a transceiver (320) and an antenna (330). The transceiver (320) receives the data send by Node S (200) through the antenna (330). The microcontroller (310) depending on the parameter warns the user and displays the water quality parameters in the LCD (340). The end-user has the provision to monitor the farm using smartphone (360) by accessing the cloud (370) or communicating with the microcontroller (310).

In an embodiment, the microcontroller (310) has the provision to upload the water quality parameters to the cloud and inform the end through user smartphone (360). The hybrid CNN-LSTM/GRU DL WQP model is implemented in the cloud (370). The WQP using DL is implemented in cloud (370) and the predicted water quality parameter can be accessed by the farmer using the smart phone via Node C (300). The microcontroller (310) may also use the GSM modem (350) to send information to the smartphone without the use of cloud (370).

Fig. 4 illustrates a flowchart (400). At step (410), initialize and sense the water quality parameters in the water body (110) using the set of sensors (230a-230d) and output the sensed signal. At step (420), the microcontroller (210) reads the sensors and stores the values in the memory. At step (430), the microcontroller (210) processes the raw data from each sensor (230a-230d) and converts to water quality parameters values. At step (440), all the final values of water quality parameters are sent to Node C (300) and the Node S (200), and at step (450), waits for defined time for collecting the water quality parameters again.

Fig. 5 illustrates the flowchart (500) for receiving the sensed data at Node C, using the deep learning hybrid CNN-LSTM/GRU model. At step (510), receiving the data from Node S (200). At step (520), the received data is stored in the database and the data is prepared for feeding to the DL model. At step (530), the data is then used for WQP using the Hybrid DL model. The Node C (300) uses the virtual machines in

cloud for running the DL models for WQP. At step (540), determine to alarm the user at step (550) or pass information in a defined time interval at step (560). The system waits for a specified time in case parameters are in the normal range at step (540), and at step (570) the process is repeated continuously passing the information to the end-user and uploading the data to the cloud.

The present invention is clearly advantageous in providing a technical solution to the problems of the prior art. The present invention is advantageous in reducing the risk of water quality issues in aquaculture systems. Further, it is also advantageous in improving the efficiency and effectiveness of aquaculture operations by providing early warning of potential water quality issues, enabling timely corrective action to be taken. This technology can be used in various types of aquaculture systems, including but not limited to marine, freshwater, and brackish water systems.

NAME OF THE APPLICANT: NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY

WE CLAIM:

1. 1. A system (100) for water quality prediction (WQP), comprising:

one or more nodal units comprises of two or more sensor nodes S (130), and a central node C (120); and

a water medium (110),

wherein the central node C (120) is connected to N number of sensor nodes S (130), each sensor node S (130a.....130n) comprises of a transceiver (220), a microcontroller unit (210) and an antenna (230) with two or more sensors (240a, 240b, 240c, and 240d), the central node C (120) comprises of a transceiver (320), a microcontroller unit (310), a display unit (340), a Global System for Mobile Communications (GSM) modem (350), an user device (360), a server (370) and an antenna (330),

wherein the sensor node S (130) is deployed in the water medium (110) to acquire water quality parameters (240a, 240b, 240c, 240d), (310) of the central node C (120) is provided for controlling and receiving data from the sensor node S (130), and the future values of water quality parameter values (240a, 240b, 240c, and 240d) is predicted from acquired water quality parameters physical sensors (240a, 240b, 240c, and 240 d), and

wherein the microcontrollers (210, 310) of the nodal units (130, 120) at the sensor node S is embedded with instructions.

2. The system (100) for water quality prediction (WQP) as claimed in claim 1, wherein the sensors acquire environmental parameters such as temperature sensor (240a), pH sensor (240b), salinity sensor (240c), dissolved oxygen (240d) and the like.

3. The system (100) for water quality prediction (WQP) as claimed in claim 1, wherein the microcontroller unit (210, 310) of the nodal units (130, 120) comprises of a memory module and an input/output module.

5. The system (100) for water quality prediction (WQP) as claimed in claim 1, wherein the memory module is configured to store the water quality parameters (240a, 240b, 240c, 240d), and the input/output module is connected to the sensors (240a, 240b, 240c, 240d) configured to receive the sensing signals.

6. The system (100) for water quality prediction (WQP) as claimed in claim 1, wherein the water quality parameters (240a, 240b, 240c, 240d) are sent to the central node C (120) from the sensor node S (130) using the transceiver (220, 320) and the antenna (230, 330), received at the central node C (120) and stored in a memory and the central node C (120) communicates the information and uploads the water quality parameters (240d, 240e) to the server (310).

7. The system (100) for water quality prediction (WQP) as claimed in claim 1, wherein the server (310) is a cloud server where all the deep learning computations occur.

8. The system (100) for water quality prediction (WQP) as claimed in claim 1, wherein the microcontroller is configured to use the GSM modem (350) to send water quality parameters (240a, 240b, 240c, 240d) information to the user device (360) by the cloud server (310) if online and by the GSM modem (350) if offline.

9. The system (100) for water quality prediction (WQP) as claimed in claim 1, wherein the user device (360) comprises of one or more laptops, mobile, smart devices or any other device.

10. A method (400a, 400b) of working of the system (100) for water quality prediction (WQP), comprising:

initialization (410) of the sensors (240a, 240b, 240c, 240d) for sensing water quality parameters in the water medium (110) and output the sensed signal;

reading (420) the sensors (240a, 240b, 240c, 240d) and storing the values in the memory by the microcontroller (210);

processing (430) the raw data from each sensor and converts to the water quality parameters and future values are predicted using deep learning model in the cloud (370) of the microcontroller (210);

sending (440) data to the central node C (120);

waiting (450) for predefined time;

receiving (510) data from the sensor node S (130) and update the data in cloud (370);

data preparation (520) is done without an error before feeding to the deep learning model;

analyzation (530) of the data using deep learning model CNN-LSTM/GRU for predicting the future values, all computations are done in the cloud (370);

checking (540) whether the predicted water quality parameters are in normal range;

wherein if the parameters are in normal range, the information is communicated (550) to the user and the data is displayed to the user device (360) by the display unit (340),

wherein if the parameters are in abnormal range, the information is communicated (560) to the user by an alarm unit,

waiting (570) for predefined time;

wherein the operation (540) is to determine to alarm the user or pass information in a defined time interval (560), and

displaying the water quality parameters values (240a, 240b, 240c, 240d) to the user device (360) by the display unit (340),

wherein the process is repeated continuously passing the information to the end-user and uploading the data to the cloud server (370).

Dated this 07th day of March 2023

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Signature: 

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HYBRID DEEP LEARNING BASED METHOD AND SYSTEM FOR AQUACULTURE WATER QUALITY PREDICTION

ABSTRACT

The present invention relates to a hybrid deep learning method and system for predicting water quality in aquaculture. The system is designed to analyse data from various sensors and instruments in real-time and uses this data to make predictions about the water quality in aquaculture. The proposed system combines Convolutional Neural Networks (CNN) and Long Short-Term Memory Networks (LSTM) or Gated Recurrent Units (GRU) to accurately predict the water quality. The proposed system is composed of two phases, including feature extraction and prediction. In the feature extraction phase, CNN is used to extract the spatio-temporal features of water quality. In the prediction phase, LSTM or GRU is used to learn the temporal patterns of water quality. The goal of the system is to improve the efficiency and effectiveness of aquaculture operations by providing early warning of potential water quality issues, enabling timely corrective action to be taken. The system is intended to be user-friendly and can be easily integrated into existing aquaculture management systems. The proposed system is tested on a real-world aquaculture dataset, and the results show that the proposed system outperforms other existing methods in terms of prediction accuracy.

Reference Figure: Fig. 2 and Fig. 3